

KNOX COUNTY REGIONAL, ME, USAWMO: **726079**Lat: **44.067N**Lon: **69.100W**Elev: **47**StdP: **14.67**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **94601****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.4	3.3	-16.8	2.2	0.2	-10.6	3.1	6.6	27.0	29.3	24.9	26.8	7.3	290	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	81.7	69.4	79.1	67.4	75.3	65.6	71.9	79.0	69.8	76.0	67.9	73.4	8.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.8	109.9	76.4	67.2	100.3	73.4	65.7	95.0	71.6	35.6	78.9	33.7	75.9	32.2	73.8	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.7	19.0	16.9	-6.4	88.4	5.8	3.2	-10.6	90.7	-13.9	92.5	-17.2	94.3	-21.3	96.6		
(5)			-7.7	75.4	5.8	2.2	-11.8	77.0	-15.3	78.2	-18.5	79.5	-22.7	81.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	22.9	25.2	32.1	41.9	51.2	59.6	66.0	65.2	58.7	48.5	39.0	29.4
	DBStd	16.55	11.17	9.57	8.26	6.04	5.85	5.88	4.78	4.57	6.12	6.70	7.79	9.04
	HDD50	3486	839	693	556	252	52	3	0	0	5	109	338	639
	HDD65	7441	1304	1113	1019	694	432	180	45	53	205	511	781	1104
	CDD50	1697	0	0	2	9	88	292	496	472	266	64	7	1
	CDD65	173	0	0	0	0	3	19	76	60	15	0	0	0
	CDH74	1035	0	0	1	1	40	150	456	297	84	6	0	0
	CDH80	178	0	0	0	0	8	27	89	41	13	0	0	0
Wind	WSAvg	7.3	8.5	8.3	8.6	8.2	7.3	6.3	5.6	5.3	5.9	7.4	7.9	8.0
Precipitation	PrecAvg	52.2	4.4	3.9	4.6	4.6	4.0	3.9	3.5	3.1	4.6	5.2	5.2	5.1
	PrecMax	73.9	8.1	8.4	9.5	10.3	9.4	9.5	7.5	7.7	11.3	15.9	9.1	8.7
	PrecMin	34.3	0.6	2.0	1.1	0.4	0.8	1.8	1.5	0.5	1.7	1.0	2.6	1.9
	PrecStd	9.8	1.9	1.3	2.0	2.4	1.9	2.1	1.6	1.9	2.7	3.2	1.6	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.7	49.6	59.5	65.9	79.7	83.7	87.7	84.3	81.3	72.0	60.9	54.5
		MCWB	50.3	45.8	49.3	52.6	64.1	70.3	73.7	73.3	68.3	61.0	55.2	51.0
	2%	DB	47.3	45.4	51.1	59.1	70.8	79.0	81.9	80.6	75.4	64.9	56.9	50.4
		MCWB	46.0	41.7	43.4	47.4	59.8	64.8	70.0	69.2	65.9	57.9	52.8	48.1
	5%	DB	43.8	42.8	46.0	54.9	65.5	73.5	79.2	77.3	72.2	62.6	54.6	46.3
		MCWB	40.6	38.9	41.0	46.3	55.8	62.7	68.2	67.0	63.8	57.1	51.1	43.7
	10%	DB	38.9	39.0	43.4	52.1	61.5	70.5	75.3	74.5	69.1	59.3	52.2	43.5
		MCWB	35.7	35.6	38.5	44.9	52.6	61.0	66.2	65.5	62.2	54.8	49.1	39.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.0	46.9	51.0	55.1	66.2	72.1	75.2	74.6	72.0	63.7	57.4	52.5
		MCDB	50.8	48.3	57.5	62.7	75.8	81.2	83.8	82.0	79.1	67.2	58.9	53.6
	2%	WB	46.1	42.9	45.7	51.1	61.3	68.3	72.1	71.5	68.2	60.9	54.3	48.3
		MCDB	46.8	45.1	48.0	55.0	68.6	76.1	79.8	77.1	72.9	63.8	56.0	49.8
	5%	WB	41.3	39.5	42.3	48.1	57.4	65.2	70.0	69.4	65.8	58.3	52.0	45.1
		MCDB	43.8	42.2	45.9	52.7	63.9	71.4	76.7	74.5	69.8	61.7	53.8	46.4
	10%	WB	36.6	36.6	39.3	46.0	54.4	62.7	68.0	67.4	63.8	55.6	49.0	40.7
		MCDB	38.7	38.9	43.0	50.2	59.4	67.6	73.8	72.3	66.5	58.9	51.9	43.8
Mean Daily Temperature Range	5% DB	MDBR	15.6	16.0	14.4	14.8	15.1	15.5	16.0	16.4	16.5	15.7	14.3	14.9
		MCDBR	17.7	16.8	17.3	20.6	23.1	21.6	20.6	20.0	20.4	18.2	15.9	18.4
		MCWBR	18.1	14.9	13.0	13.0	13.6	12.5	11.4	11.8	12.8	13.1	14.1	17.7
	5% WB	MCDBR	17.8	16.2	16.0	16.6	19.9	19.2	17.9	17.2	16.5	15.8	14.2	18.0
		MCWBR	18.5	15.5	13.3	12.5	13.3	12.9	11.2	11.2	11.6	13.2	14.1	18.7
Clear-Sky Solar Irradiance	taub		0.297	0.305	0.328	0.354	0.380	0.422	0.441	0.410	0.367	0.338	0.319	0.298
	taud		2.469	2.457	2.461	2.425	2.396	2.291	2.260	2.336	2.437	2.518	2.535	2.515
	Ebn at Noon		265	283	290	289	283	270	263	268	272	267	255	254
	Edh at Noon		23	27	31	35	37	41	42	37	31	25	21	20
All-Sky Solar Radiation	RadAvg		506	790	1171	1463	1660	1787	1872	1687	1358	871	543	400
	RadStd		38	63	73	155	200	150	135	89	81	80	49	42